

IEEE IES Generative AI Challenge 2026

Mentorship Introduction & Project Overview

Team Introduction

We are a two-member team participating in the IEEE IES Generative AI Challenge 2026. Below is a brief introduction to each team member.

Dimpana P Jadhav [Portfolio](#)

Institution: BNM Institute of Technology, Bengaluru - B.E. Mechanical Engineering (2023–2027), CGPA 9.14

Background: Aerospace-oriented design, simulation, and mechanical system development.

Current Role: Product Development Intern, Autoyos Pvt. Ltd. mechanical design, prototyping, and product development

Key experience:

- Aerodynamic design and CFD analysis of a fixed-wing UAV (Vayudhwajai-M Aerospace internship)
- CAD modelling and structural integration for a medical logistics drone (DRONE–SANJEEVINI project)
- Winner – Shikshalokam Hackathon 2026 (Just-in-Time Classroom AI agent)

Tanmay Anand [LinkedIn](#)

Institution: BNM Institute of Technology, Bengaluru - B.E. Computer Science & Engineering (2023–2027), GPA 9.0

Background: Machine learning, backend development, and DevOps

Internship: Software Engineering Intern, BETSOL focusing on FastAPI backend development, RBAC with Azure Entra ID, MySQL + Redis optimisation

Key experience:

- Fine-tuned BERT/LSTM models for NLP classification (IEEE internship) - F1-score 0.87 on a 50,000-sample dataset
- Real-time licence plate recognition using YOLOv8/EasyOCR (ALPERS project, 94% accuracy, deployed in residential complexes)
- Multimodal threat detection with Gemini 2.5 Flash (Bharat Raksha, Thales Hackathon Finalist)
- Winner - Shikshalokam Hackathon 2026 (Just-in-Time Classroom AI agent)

Highlights of the Proposed Project

1. Project Overview and Depth

Aircraft turbofan engines undergo progressive degradation over their operational lifecycle. Predicting their Remaining Useful Life (RUL) accurately is critical for preventing unplanned failures, minimizing downtime, and optimizing maintenance costs. The current state of the art predominantly delivers single-point RUL estimates - a number with no uncertainty bounds, no

information about trajectory variability, and no indication of whether the prediction is physically plausible.

Our project — Physics-Integrated Generative Edge-AI Framework for Explainable Aero-Engine Prognostics — addresses this gap by replacing scalar point prediction with generative counterfactual prognostics. The system:

- Encodes multi-variate sensor time-series windows using an attention-based architecture trained on the NASA C-MAPSS FD001 benchmark
- Feeds latent degradation embeddings into a physics-constrained generative model (conditional VAE or normalizing flow) to sample multiple plausible future degradation trajectories
- Evaluates each generated trajectory against physics-derived constraints such as monotonic degradation and thermodynamic consistency
- Produces a dual-head output: (A) a probabilistic RUL distribution and (B) a physics-violation risk score per trajectory
- Surfaces all outputs through a reliability gating layer (ACCEPT / WARN / REJECT) accessible via a Streamlit dashboard, FastAPI REST endpoints, and an MQTT-connected live digital twin interface

A working prototype already supports Baseline attention model inference, a Physics-Informed model variant, and live MQTT digital twin ingestion, all validated on C-MAPSS FD001.

2. Generative AI Application

Generative AI is not a peripheral component in this system but it is the core of the prognostic pipeline. After the attention encoder produces a latent degradation state vector from a sliding sensor window, a physics-constrained generative model samples N plausible future degradation trajectories conditioned on that state.

These trajectories serve two purposes simultaneously:

- **Head A (RUL Distribution):** The endpoint distributions of sampled trajectories are aggregated to yield a mean RUL estimate, a confidence interval, and a variance measure, replacing the scalar output of conventional approaches.
- **Head B (Physics-Violation Risk):** Each generated trajectory is scored against learned physics constraints embedded in the model's loss function. Trajectories that violate thermodynamic or degradation-monotonicity priors are flagged, and the aggregate violation rate becomes the physics-violation risk score.

This generative stage also enables counterfactual reasoning: the system can answer what-if questions such as 'What would RUL look like if operating temperatures were elevated by 5%?' - a capability entirely absent from point-prediction approaches.

3. Data Acquisition and Quality Assurance

The primary dataset is the NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) FD001 subset - a well-established public benchmark consisting of 100 training engines and 100 test engines operating under a single fault mode and operating condition, with 21 sensor channels and 3 operational settings recorded per cycle.

Our data quality pipeline applies the following strategies:

- **Near-zero-variance sensor removal** identified through ablation analysis, reducing input dimensionality to the most informative channels.
- **Min-max normalization** fitted exclusively on training splits to prevent data leakage into validation and test sets.

- **Sliding-window construction** with controlled stride and validated window boundaries.
- Piecewise-linear RUL label **capping at 125 cycles**, consistent with the established literature for this benchmark.
- A validation suite with regression tests against frozen golden outputs to detect preprocessing drift across development iterations.

If the project is extended to multi-condition subsets (FD002-FD004) or real industrial sensor streams, the preprocessing pipeline is designed to accommodate configurable feature mappings and normalization profiles per operating regime.

4. Integration with Industrial Systems

The system architecture is designed around modular, interface-first boundaries to minimize integration friction with existing industrial ecosystems:

- **FastAPI REST backend:** Exposes standardized inference endpoints callable from SCADA systems, MES platforms, or maintenance dashboards over HTTP, no proprietary integration required.
- **MQTT ingestion layer:** Compatible with Mosquitto brokers and supports TLS-secured connections, enabling direct integration with industrial IoT sensor networks. A digital twin streamer also allows integration testing without physical hardware.
- **Edge export scripts:** The attention model can be quantized and exported for deployment on resource-constrained edge devices, supporting on-device inference near the sensor source.
- **Streamlit dashboard:** A human-readable operational interface with real-time monitoring, batch inference, and replay capabilities for MRO operators.

Anticipated integration challenges include: sensor sampling rate mismatches between C-MAPSS simulated data and real engine telemetry; network reliability constraints in factory or hangar environments; and the need to map proprietary sensor schema definitions to the C-MAPSS feature space. We plan to address these through configurable preprocessing adapters with per-deployment schema mapping configurations.

5. Innovation and Differentiation

The key technical differentiators from existing literature are:

- **Counterfactual prognostics over point prediction:** Existing systems output a single RUL value. Our system generates a distribution of physically grounded future trajectories, enabling uncertainty quantification, scenario analysis, and operator confidence calibration.
- **Embedded physics constraints, not post-hoc filtering:** Physics-derived constraints are encoded directly into the generative model's training loss ensuring generated trajectories are physically coherent by construction, not filtered retrospectively.
- **Counterfactual Physical Consistency (CPC) metric:** A novel explainability metric we are formalizing to quantify whether counterfactual trajectories produced by the model remain consistent with known engine physics. This provides a domain-grounded interpretability signal that goes beyond standard attribution methods like SHAP or LIME.
- **Dual-head risk output:** Operators receive both a probabilistic RUL estimate and an explicit physics-violation risk flag enabling more defensible, actionable maintenance scheduling decisions.

Prior work has largely treated prediction and explainability as sequential steps. Our contribution integrates generative counterfactual reasoning with embedded physics validity checking, producing a system that is simultaneously predictive, explainable, and physically accountable.

6. Challenges and Solutions

Physics constraint formulation: Translating engine domain knowledge such as degradation monotonicity and thermodynamic bounds into differentiable loss terms is non-trivial. We plan to begin with soft penalty terms applied to sensor trajectory gradients and iteratively tighten constraints based on validation feedback and mentor guidance.

Generative model training stability: Conditional VAEs are susceptible to posterior collapse, where the decoder learns to ignore the latent code. We will apply KL annealing schedules and monitor latent space utilization metrics throughout training to ensure the generative component remains expressive.

Edge deployment of the generative component: Sampling-based inference is compute-intensive. We will profile the full pipeline on target hardware early in development and apply trajectory pruning or deterministic approximation strategies if latency budgets are exceeded.

Benchmark comparability: Ensuring fair comparison against the Baseline attention model and the LightGBM notebook artifact requires frozen preprocessing pipelines and evaluation splits enforced via our existing validation suite and regression test scaffold.

7. Expectations from the Mentor

We would most value mentorship in the following specific areas:

- **Physics-ML integration:** Guidance on best practices for embedding thermodynamic or mechanical degradation physics as training constraints, in particular, whether soft penalties, hard projection layers, or physics-informed loss architectures are most appropriate for the C-MAPSS degradation regime.
- **Generative model architecture selection:** Insight into whether a conditional VAE, normalizing flow, or diffusion-based model is best suited to the time-series structure of C-MAPSS data and the edge deployment constraints we face.
- **Research positioning and gap identification:** Guidance on framing the CPC metric contribution rigorously for competition evaluation and identifying the most defensible research gaps in the current aero-engine prognostics literature.
- **Industrial credibility:** Any perspective on how prognostic systems are actually evaluated or adopted in MRO (Maintenance, Repair, Overhaul) operational contexts to help ground our claims and ensure the system design reflects real-world usability requirements.

We are happy to share our current codebase, model outputs, or validation summaries at your convenience.

Thank you for your time, and we look forward to your mentorship.

Dimpana P Jadhav & Tanmay Anand.